

# Class XI

## SUBJECT : MODERN COMPUTER APPLICATION(COMA)

### SEMESTER – I

**Total Contact Hours: 200 (Theory & Practical: 180 ; Remedial & Home Assignment: 20)**

**Course Code : COMA (Theory)**

**Full Marks : 35**

**Contact Hours : 100**

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| Unit – 1 | <b>Computer System and Organisation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>15 Marks</b> | <b>Total 30 Hours</b> |
|          | <ul style="list-style-type: none"><li>• Basic Computer Organisation<ul style="list-style-type: none"><li>➤ CPU, Primary Memory (RAM, ROM, Cache), Secondary storage device, I/O devices, units of memory (bit, byte, KB, MB, GB, TB, PB).</li></ul></li><li>• Classification of Computers<ul style="list-style-type: none"><li>➤ Super, Mainframe, Mini, PC.</li></ul></li></ul>                                                                                                                                        |                 | <b>4 Hours</b>        |
|          | <ul style="list-style-type: none"><li>• Concepts of Software<ul style="list-style-type: none"><li>➤ Definition of software, types of software – System Software (Translator: assembler, interpreter, compiler, Loader, Linker, Operating System: Definition and functions, types of OS- Single use, Multiuse, Multiprogramming, Multiprocessing, Time sharing), Application Software (Definition and example), Utility Software, concept of GUI and CUI with examples using LINUX (Basic Commands).</li></ul></li></ul> |                 | <b>7 Hours</b>        |
|          | <ul style="list-style-type: none"><li>• Number System<ul style="list-style-type: none"><li>➤ Binary, Octal, Decimal, Hexadecimal number system, conversion between number system, Weighted Code (BCD, Binary, 84-2-1 code), non-weighted code (GREY, Excess-3), encoding schemes (ASCII, ISCII, uni code), 1's complement, 2's complement.</li></ul></li></ul>                                                                                                                                                          |                 | <b>9 Hours</b>        |
|          | <ul style="list-style-type: none"><li>• Boolean Algebra<ul style="list-style-type: none"><li>➤ Postulates, logic gates: NOT, AND, OR, NAND, XOR, NOR, XNOR, truth tables, De Morgan theorem, SOP, POS, Simplifications using KMap and Boolean algebra, logic circuits.</li></ul></li></ul>                                                                                                                                                                                                                              |                 | <b>10 Hours</b>       |
| Unit – 2 | <b>Programming Fundamentals</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>10 Marks</b> | <b>Total 20 Hours</b> |
|          | <ul style="list-style-type: none"><li>• Concept of Programming<ul style="list-style-type: none"><li>➤ Instruction (Definition, Example), Program (definition, example), Programming Language (concept of high level, low level and assembly language), Procedural and Non-procedural programming, Concept of Structured Programming, Object Oriented Programming.</li></ul></li></ul>                                                                                                                                   |                 | <b>2 Hours</b>        |
|          | <ul style="list-style-type: none"><li>• Algorithm fundamentals<ul style="list-style-type: none"><li>➤ Definition, characteristic of algorithm, recursive and non-recursive algorithms, representation of algorithm using flowchart, pseudo code.</li></ul></li></ul>                                                                                                                                                                                                                                                    |                 | <b>15 Hours</b>       |
|          | <ul style="list-style-type: none"><li>• Introduction to Problem Solving<ul style="list-style-type: none"><li>➤ Steps for Problem Solving (analysing the problem, developing an algorithm, coding, testing, debugging).</li></ul></li></ul>                                                                                                                                                                                                                                                                              |                 | <b>3 Hours</b>        |

| Unit – 3 | Data Visualization using Spreadsheet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10 Marks | Total 50 Hours |
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|          | <ul style="list-style-type: none"> <li>Introduction to Spreadsheets               <ul style="list-style-type: none"> <li>➤ Spreadsheets and their applications, overview of spreadsheet software (e.g., Open office, Google Sheets, Excel), creating workbooks, modifying workbook, zooming in on a worksheet, arranging multiple workbook windows, adding buttons to the quick access toolbar, customizing the ribbon, maximizing usable space in the program window navigating the spreadsheet interface, entering and editing data in cells saving, opening, and closing spreadsheet files.</li> </ul> </li> </ul> |          | 6 Hours        |
|          | <ul style="list-style-type: none"> <li>Working with Data and Tables               <ul style="list-style-type: none"> <li>➤ Entering and revising data, moving data within a workbook, finding and replacing data, correcting and expanding upon worksheet data, defining tables.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                               |          | 5 Hours        |
|          | <ul style="list-style-type: none"> <li>Performing Calculations on Data               <ul style="list-style-type: none"> <li>➤ Naming groups of data, creating formulas to calculate values (e.g., SUM, AVERAGE, COUNT), summarizing data that meets specific conditions (e.g., AVERAGEIF, COUNTA, COUNTBLANK, COUNTIFS, SUMIF, IFERROR etc), finding and correcting errors in calculations.</li> </ul> </li> </ul>                                                                                                                                                                                                    |          | 5 Hours        |
|          | <ul style="list-style-type: none"> <li>Changing Workbook Appearance               <ul style="list-style-type: none"> <li>➤ Formatting Cells, defining styles, workbook themes and table styles, making numbers easier to read, changing the appearance of data based on its value, adding images to worksheets.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                |          | 4 Hours        |
|          | <ul style="list-style-type: none"> <li>Data Analysis and Manipulation               <ul style="list-style-type: none"> <li>➤ Limiting data appearance on screen, working with text functions for data cleaning, Splitting and combining data, Data normalization and standardization, working with ranges and named ranges, conditional formatting, data validation and error checking, using logical functions (e.g., IF, AND, OR), sorting and filtering data.</li> </ul> </li> </ul>                                                                                                                               |          | 10 Hours       |
|          | <ul style="list-style-type: none"> <li>Advanced Spreadsheet Features               <ul style="list-style-type: none"> <li>➤ Creating and managing tables, working with charts and graphs.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                      |          | 10 Hours       |
|          | <ul style="list-style-type: none"> <li>Reporting and Presentation of Results               <ul style="list-style-type: none"> <li>➤ Designing informative reports and summaries, creating interactive dashboards for data presentation, data visualization best practices, documenting data analysis processes presenting findings to stake holders.</li> </ul> </li> </ul>                                                                                                                                                                                                                                           |          | 8 Hours        |
|          | <ul style="list-style-type: none"> <li>Collaboration and Sharing               <ul style="list-style-type: none"> <li>➤ Protecting worksheets and workbooks, sharing spreadsheets with others, tracking changes and commenting,</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                |          | 2 Hours        |

**Class XI**  
**SUBJECT : MODERN COMPUTER APPLICATION(COMA)**  
**SEMESTER – II**

**Course Code: COMA (Theory)**

**Full Marks : 35**

**Contact Hours : 80**

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| Unit – 1 | <b>Data Structure</b>                                                                                                                                                                                                                                                                                                                                                                                                    | <b>10 Marks</b> | <b>Total 30 Hours</b> |
|          | <ul style="list-style-type: none"> <li>Definition, types of data structure-linear and non-linear.</li> </ul>                                                                                                                                                                                                                                                                                                             |                 | <b>2 Hours</b>        |
|          | <ul style="list-style-type: none"> <li>Arrays: 1D, 2D and their applications.</li> </ul>                                                                                                                                                                                                                                                                                                                                 |                 | <b>3 Hours</b>        |
|          | <ul style="list-style-type: none"> <li>Linked List: Basic concepts of Single, circular and double link list.</li> </ul>                                                                                                                                                                                                                                                                                                  |                 | <b>6 Hours</b>        |
|          | <ul style="list-style-type: none"> <li>Stack <ul style="list-style-type: none"> <li>Stack operations (push and pop), applications of Stack.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                       |                 | <b>4 Hours</b>        |
|          | <ul style="list-style-type: none"> <li>Queue <ul style="list-style-type: none"> <li>Queue operations, applications of queue, basic concepts of circular queue and priority queue.</li> </ul> </li> </ul>                                                                                                                                                                                                                 |                 | <b>4 Hours</b>        |
|          | <ul style="list-style-type: none"> <li>Recursion <ul style="list-style-type: none"> <li>Definition.</li> <li>Advantages and limitations of recursion.</li> </ul> </li> </ul>                                                                                                                                                                                                                                             |                 | <b>4 Hours</b>        |
|          | <ul style="list-style-type: none"> <li>Searching and Sorting <ul style="list-style-type: none"> <li>Linear Search, Binary Search, and their comparison.</li> <li>Bubble Sort and its Implementation.</li> </ul> </li> </ul>                                                                                                                                                                                              |                 | <b>7 Hours</b>        |
| Unit – 2 | <b>Computer Networks</b>                                                                                                                                                                                                                                                                                                                                                                                                 | <b>15 Marks</b> | <b>Total 35 Hours</b> |
|          | <b>&gt;Introduction to Networking</b><br>>>Analogue and digital Communication<br>>>Mode of Communication- Simplex, half duplex and full duplex<br>>>Network Architecture- Client server, Peer to Peer<br>>>Serial and Parallel Communication<br>>>Measuring Capacity of Communication Media (bandwidth, channel capacity, baud)<br>>>Synchronous and asynchronous Transmission Mode<br>>>Baseband and Broadband network. |                 | <b>6 Hours</b>        |
|          | <b>&gt;Transmission Media</b><br>>>Wired Communication Media (Twisted Pair, Co-axial cable, Fiber Optic).<br>>>Wireless Communication Media (Radio wave, Microwave, Infrared, Satellite).                                                                                                                                                                                                                                |                 | <b>3 Hours</b>        |
|          | <b>&gt;Network Connecting Devices</b><br>>>Modem, Ethernet Card, RJ45, Repeater, Hub, Switch, Router, Gateway, Wifi card.                                                                                                                                                                                                                                                                                                |                 | <b>2 Hours</b>        |

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|          | <b>&gt;Network Type and Topologies</b><br>>>Types of Network-LAN, MAN, WAN.<br>>>Network Topologies- Bus, Star, Ring, Tree.                                                                                                                                                                                                 | <b>3 Hours</b>  |
|          | <b>&gt;Network Protocols</b> -HTTP, FTP, PPP, SMTP, TCP/IP, POP3, TELNET, HTTPS, VoIP.                                                                                                                                                                                                                                      | <b>2 Hours</b>  |
|          | <b>&gt;Referential Model-</b> OSI Model (Basic Concept, use of devices and protocols at different layers).                                                                                                                                                                                                                  | <b>1 Hours</b>  |
|          | <b>&gt;Introduction to Web Services:</b> WWW, HTML, XML, IP Addresses, Domain names, URL, ISP, Website, Web browser, Web Server, Web Hosting.                                                                                                                                                                               | <b>3 Hours</b>  |
|          | <b>&gt;HTML</b><br>Basic Tags and Document structure, HTML Tags, Head Tags, Title Tags, Introduction to HTML and Web design, how to create simple Web page, how to format text, Create Table, Adding Web link and Images, Forms, adding styles and classes to web pages, Borders and Background, Adding Video and Graphics. | <b>15 hours</b> |
| Unit – 3 | <b>Ethics</b>                                                                                                                                                                                                                                                                                                               | <b>10 Marks</b> |
|          | <ul style="list-style-type: none"> <li><b>Digital Footprints.</b></li> </ul>                                                                                                                                                                                                                                                | <b>1 Hour</b>   |
|          | <ul style="list-style-type: none"> <li><b>Data Protection:</b> Intellectual property rights (copyright, patent, trademark), violation of IPR (plagiarism, copyright infringement, trademark infringement), open-source software and licensing (Creative Commons, GPL and Apache).</li> </ul>                                | <b>5 Hours</b>  |
|          | <ul style="list-style-type: none"> <li><b>Cyber Crime:</b> Definition, hacking, eavesdropping, phishing and fraud emails, ransomware, cyber trolls, cyber bullying.</li> </ul>                                                                                                                                              | <b>3 Hours</b>  |
|          | <ul style="list-style-type: none"> <li><b>Cyber safety:</b> Safely browsing the web, identity protection, confidentiality.</li> </ul>                                                                                                                                                                                       | <b>2 Hours</b>  |
|          | <ul style="list-style-type: none"> <li><b>Malware:</b> Viruses, trojans, adware.</li> </ul>                                                                                                                                                                                                                                 | <b>1 Hour</b>   |
|          | <ul style="list-style-type: none"> <li><b>E-waste management:</b> Proper disposal of used electronic gadgets.</li> </ul>                                                                                                                                                                                                    | <b>2 Hours</b>  |
|          | <ul style="list-style-type: none"> <li><b>Information Technology Act :</b> (IT Act).</li> </ul>                                                                                                                                                                                                                             | <b>1 Hour</b>   |